

Program 8

Create Kubernetes Pods using YAML Descriptors

J Ryan Roy (1RV24MC045)

Steps:

1. Create a nodejs application

Inside server.js

```
const http = require("http");

const port = 3000;

const server = http.createServer((req, res) => {
  res.end("Hello from Node.js running inside Kubernetes");
});

server.listen(port, () => {
  console.log(`Server running at port ${port}`);
});
```

2. Create a configmap yaml file to create a nodejs pod

Inside nodejs-configmap.yaml

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");

    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
```

3. Create a pods yaml file to configure pods

Inside nodejs-pods.yaml

```
apiVersion: v1
kind: Pod
metadata:
  name: program-8
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      ports:
        - containerPort: 3000
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

4. Apply Both yaml files

```
(1rv24mc045_ryan@ryan-kali) ~ - [~/Documents/program8]
$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config created

(1rv24mc045_ryan@ryan-kali) ~ - [~/Documents/program8]
$ kubectl apply -f nodejs-pods.yaml
pod/program-8 created
```

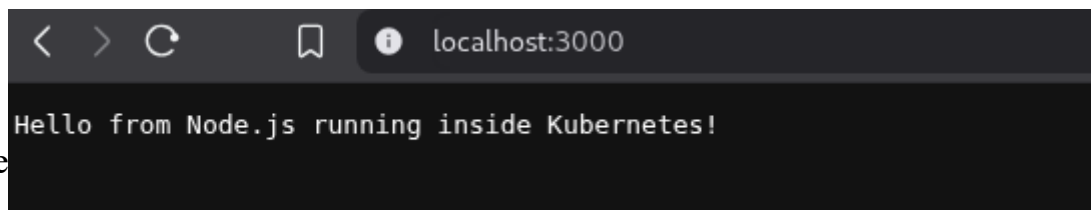
5. Check pods to see if created.

```
(1rv24mc045_ryan@ryan-kali) ~ - [~/Documents/program8]
$ kubectl get pods
NAME      READY STATUS  RESTARTS AGE
program-8 1/1   Running 0       3m23s
```

6. Enable port forwarding so that the pod is accessible.

```
(1rv24mc045_ryan@ ryan-kali) ~/Documents/program8
$ kubectl port-forward pod/program-8 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
```

7. Access the pod at <http://localhost:3000>



8.
Delete
all
pods.

```
(1rv24mc045_ryan@ ryan-kali) ~/Documents/program8
$ kubectl delete pods --all
pod "program-8" deleted
```